

ORF307

Least squares data fitting

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Today's topics

- Least squares model fitting
- Regression in one variable
- Multivariate regression
- Validation
- Example

Least squares model fitting

Problem data and set up

- We have a **dataset** of some measurements (aka, observations or examples)

$$D = \{(x^{(i)}, y^{(i)}) : i = 1, 2, \dots, N\}, \quad x^{(i)} \in R^n, \quad y^i \in R$$

- ❖ The **j -th element** of $x^{(i)}$ will be denoted by $x_j^{(i)} \in R$
- ❖ The $x_j^{(i)}$'s are called **features**
- ❖ The $y^{(i)}$'s are called the **ground truth**
- The relation between the $x^{(i)}$'s and the $y^{(i)}$'s is **governed by an unknown function** $f: R^n \rightarrow R$
- **Goal 1**: Estimate (approximate/learn/discover) the function f from the available dataset D

The Model

- **Discover** a function $\hat{f}: R^n \rightarrow R$ that **approximates** f

$$\hat{f}(x) = \theta_1 f_1(x) + \theta_2 f_2(x) + \cdots + \theta_p f_p(x)$$

- The functions $f_i: R^n \rightarrow R$ are called **basis functions** (or feature mappings)
 - You have the **freedom** to define them as you like
- The θ_i 's are the **model parameters (unknown)**
- $\hat{y}^{(i)} = \hat{f}(x^{(i)})$ is the **model's prediction** for the ground truth $y^{(i)}$
- **Goal 2:** you want the $\hat{y}^{(i)}$'s to be **as close to** $y^{(i)}$ **as possible** ($\hat{y}^{(i)} \approx y^{(i)}$)

Least squares data fitting

- Define the **prediction errors** (aka, **residuals**) as: $r^{(i)} = y^{(i)} - \hat{y}^{(i)}$
- **Equivalent Goal 2:** choose the model parameters θ_i such that **the Mean Squared Error (MSE) is minimized**

$$\frac{(r^{(1)})^2 + (r^{(2)})^2 + \dots + (r^{(N)})^2}{N}$$

- This can be expressed as a **Least Squares problem**
 - Let's see how...

Least squares data fitting

- Define the vectors:
- $y^d = (y^{(1)}, y^{(2)}, \dots, y^{(N)})^T$ **outcomes (ground truth)**
- $\hat{y}^d = (\hat{y}^{(1)}, \hat{y}^{(2)}, \dots, \hat{y}^{(N)})^T$ **predictions**
 - $\hat{y}^{(i)} = \hat{f}(x^{(i)}) = \theta_1 f_1(x^{(i)}) + \dots + \theta_p f_p(x^{(i)})$
 - Set: $A_{ij} = f_j(x^{(i)})$ (**note**: you define the f_j 's)
 - $\hat{y}^{(i)} = \hat{f}(x^{(i)}) = \theta_1 A_{i1} + \dots + \theta_p A_{ip}$
 - $\hat{y}^d = A\theta$, where $\theta = (\theta_1, \dots, \theta_p)^T$

- $r^d = (r^{(1)}, r^{(2)}, \dots, r^{(N)})^T$ **residuals**

- $r^{(i)} = y^{(i)} - \hat{y}^{(i)}, i = 1, \dots, N$

- $r^d = y^d - \hat{y}^d \Rightarrow r^d = y^d - A\theta$

- **Least squares problem:**

$$\min \|r^d\|^2 = \min \|y^d - A\theta\|^2$$

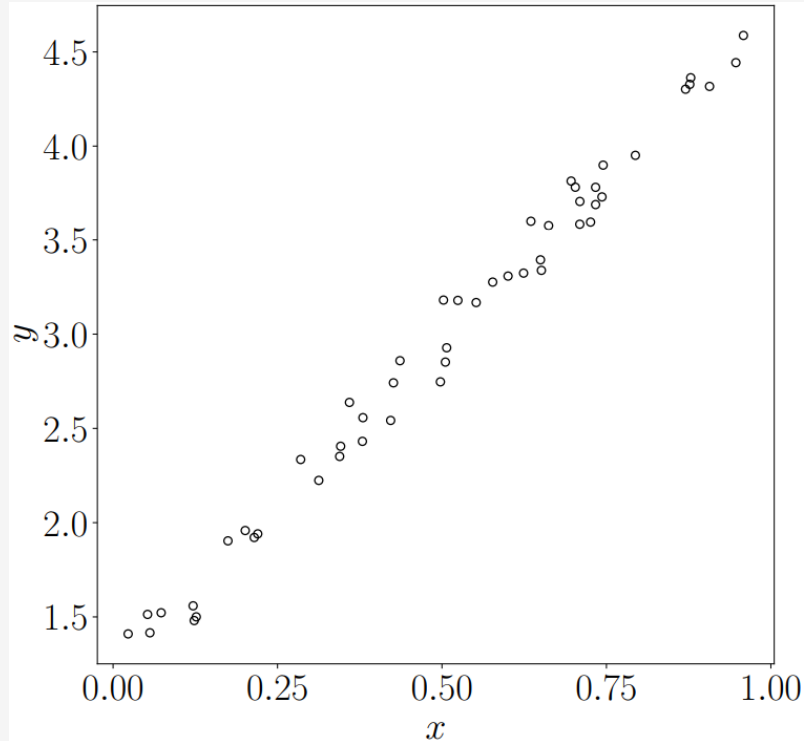
- **Solution (normal equations):**

$$A^T A \theta^* = A^T y^d$$

Univariate fitting

Fitting *univariate* functions

- We have the **dataset**:
- $D = \{(x^{(i)}, y^{(i)}) : i = 1, 2, \dots, N\}$, $x^{(i)} \in \mathbf{R}$, $y^{(i)} \in R$



- **Our fitting model is:** $\hat{f}(x) = \theta_1 + \theta_2 x$
- Parameters: $\theta = (\theta_1, \theta_2)^T \in R^2$
- Feature mappings: $f_1(x) = 1$, $f_2(x) = x$
- The matrix
$$A = \begin{bmatrix} 1 & x^{(1)} \\ \vdots & \vdots \\ 1 & x^{(N)} \end{bmatrix} = [1 \ x^d]$$
- **Goal:** find θ_1, θ_2 such that

$$\min \|r^d\|^2 = \|y^d - A\theta\|^2$$

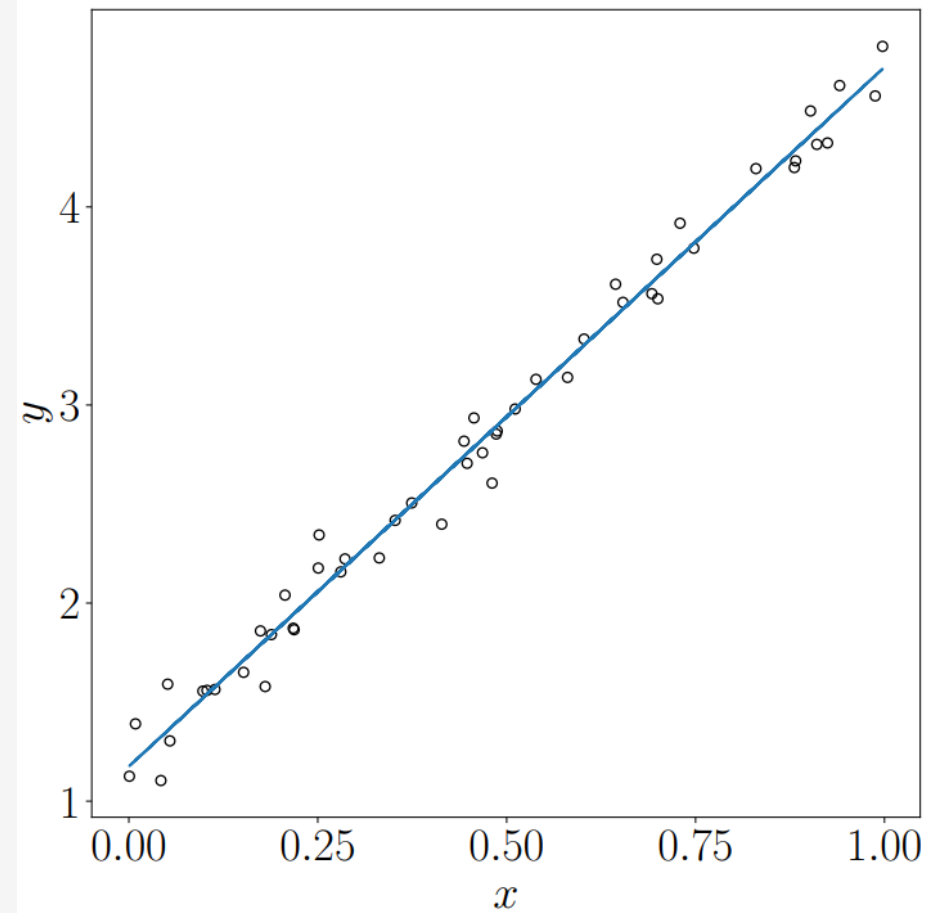
Fitting univariate functions

- **Example:**

- It is just a straight line

- $\theta^* = (1.75, 3.53)^T$

- $y \approx \hat{f}(x) = 1.75 + 3.53x$



Example #1: Asset α and β in finance (generate α)

- Assume you have your own asset (e.g., portfolio of stocks)
- **Your Goal:** predict your asset's returns from total market returns
- **Data** (going back t days):
 - $x^d = (r_1^{mkt}, \dots, r_t^{mkt})^T$: total market returns
 - $y^d = (r_1^{ind}, \dots, r_t^{ind})^T$: individual asset returns
- **Linear regression model:** $\hat{f}(x) := \hat{y} = (r^{rf} + \alpha) + \beta(x - \mu^{mkt})$
 - r^{rf} : risk-free interest rate,
 - $\mu^{mkt} = avg(x^d)$: average market return.
- **Unknown parameters:** α and β
 - $\theta_1 = r^{rf} + \alpha - \beta\mu^{mkt}$ and $\theta_2 = \beta$
 - α : average asset return above the risk-free interest rate
 - β : systematic risk of the individual asset relative to the overall market.
 - $\beta = 1$: as volatile as the market
 - $\beta > 1$: more volatile than market
 - $\beta < 1$: less volatile than market

Example #2: time series trend

- You have recorded the values of some phenomenon at N time points

- Your goal:** estimate the trend

- Data:**

- $x^d = (1, \dots, N)^T$ and $y^d = (y^{(1)}, \dots, y^{(N)})^T$

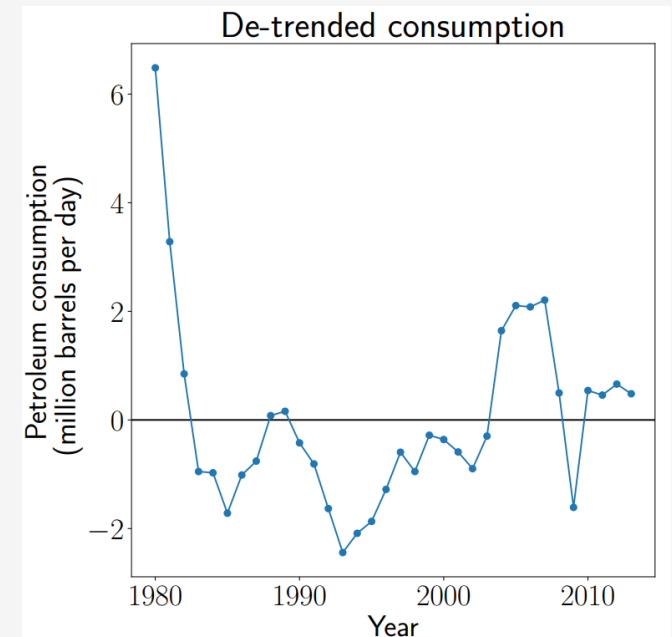
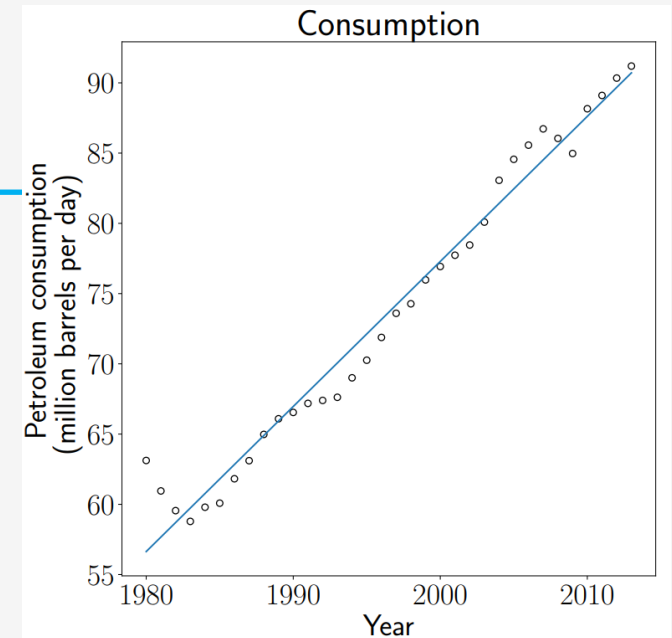
- Linear regression model:** $\hat{f}(x) := \hat{y} = \theta_1 \mathbf{1} + \theta_2 x^d$

- That is: $\hat{y}^{(i)} = \theta_1 \mathbf{1} + \theta_2 i$ for $i = 1, 2, \dots, N$

- θ_2 : trend coefficient

- $y^d - \hat{y}$: **de-trending time series**

Helps us identify seasonality or cyclical behavior



Example #3: Polynomial fit

- You now want to consider **more than two feature functions**

$$f_i(x) = x^{i-1}, \quad \text{for } i = 1, 2, 3, \dots, p$$

- Your regression model becomes:

$$\hat{f}(x) := \hat{y} = \theta_1 + \theta_2 x + \theta_3 x^2 + \dots + \theta_p x^{p-1} = A\theta$$

$$\text{where, } A = \begin{bmatrix} 1 & x^{(1)} & (x^{(1)})^2 & \dots & (x^{(1)})^{p-1} \\ 1 & x^{(2)} & (x^{(2)})^2 & \dots & (x^{(2)})^{p-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x^{(N)} & (x^{(N)})^2 & \dots & (x^{(N)})^{p-1} \end{bmatrix} \text{ and } \theta = \begin{pmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \\ \vdots \\ \theta_p \end{pmatrix}$$

Notation:

x^i : the i -th power

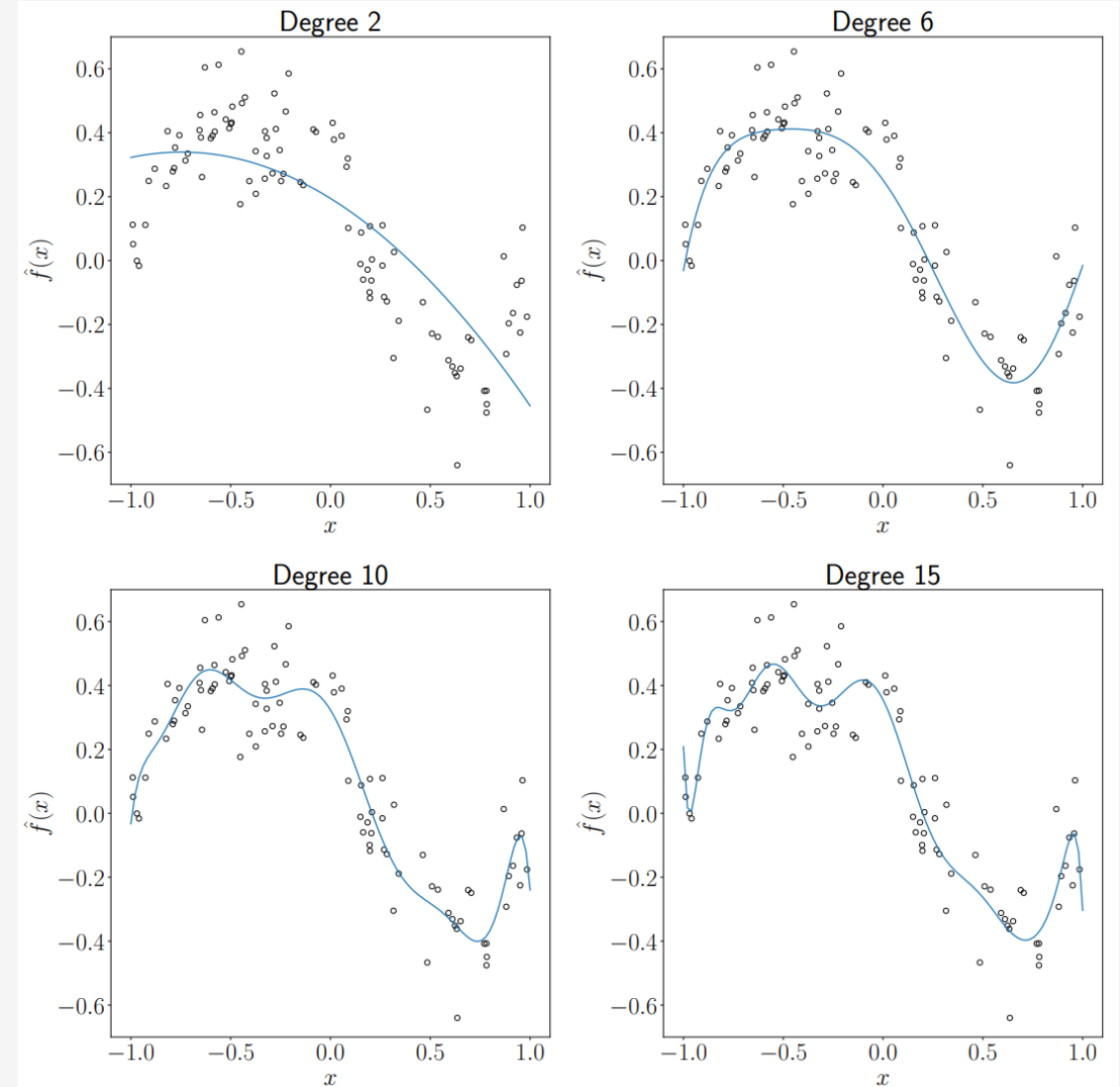
$x^{(i)}$: the i -th data point

Goal:

$$\min ||r^d||^2 = ||y^d - A\theta||^2$$

Example #3: Polynomial fit

- When we have $N = 100$ data points
- Which model is better predictor?



Example #4: Autoregressive time series models

- You have collected time series data (e.g., stock prices, temperatures, etc.)

$$z_1, z_2, z_3, \dots, z_N$$

- An **autoregressive (AR) prediction model** is defined as

M : the **memory** or **lag** of the model

$$\hat{z}_{t+1} = \theta_1 z_t + \theta_2 z_{t-1} + \dots + \theta_M z_{t-M+1}, \quad \text{for } t = M, M+1, \dots, N$$

- Goal**: choose the best θ 's by solving the optimization problem

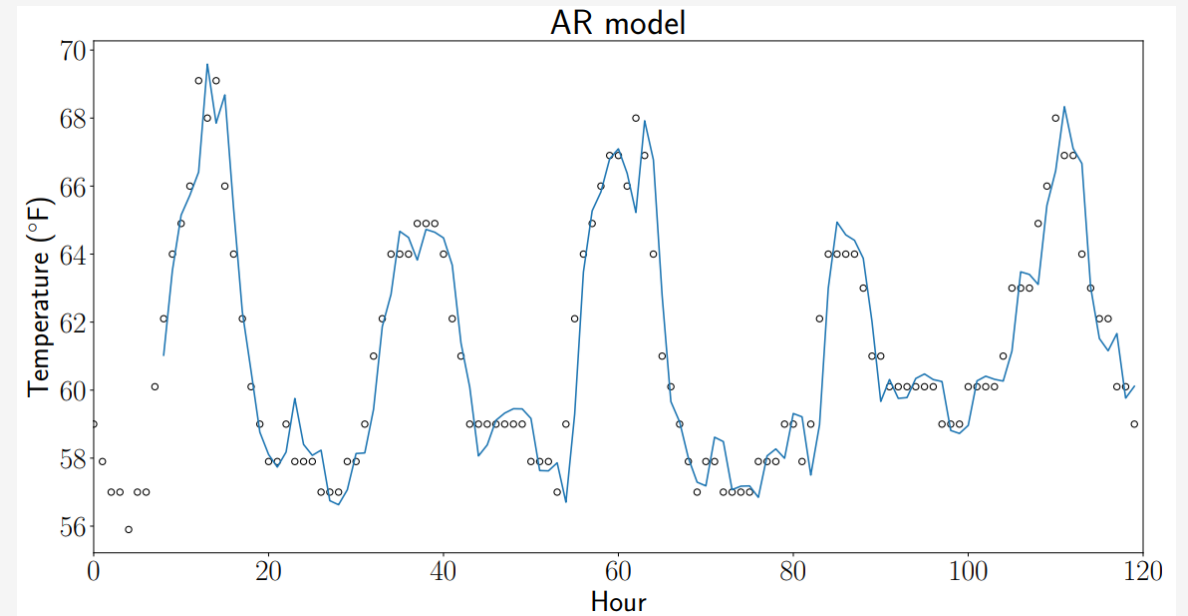
$$\min (\hat{z}_{M+1} - z_{M+1})^2 + (\hat{z}_{M+2} - z_{M+2})^2 + \dots + (\hat{z}_N - z_N)^2$$

- Remark**: the model can be converted to standard linear regression by defining:

- $y^{(i)} = z_{M+i}$ and $x^{(i)} = (z_{M+i-1}, z_{M+i-2}, \dots, z_i)^T$, for $i = 1, 2, \dots, N - M$ ($\rightarrow$ **Exercise!**)

Example #4: Autoregressive time series models

- Assume you have hourly temperature data for **LAX** in May 2016, i.e., $N = 31 * 24$
- We fit an AR model with **lag** $M = 8$
 - The **RMS Error** is: $0.98^{\circ}F$
- **Comparison with other models:**
 - Previous hour: $\hat{z}_{t+1} = z_t \rightarrow \text{RMSE} = 1.16^{\circ}F$
 - Same hour, but yesterday: $\hat{z}_{t+1} = z_t \rightarrow \text{RMSE} = 1.73^{\circ}F$



Multivariate regression

Linear regression as general data fitting model

- The **standard linear regression model** is defined as

$$\hat{y} = \hat{f}(x) = x^T \beta + v = x_1 \beta_1 + x_2 \beta_2 + \cdots + x_{n-1} \beta_{n-1} + v$$



- General **data fitting model**

$$\hat{f}(x) = \theta_1 f_1(x) + \theta_2 f_2(x) + \cdots + \theta_n f_n(x)$$

- With basis functions: $f_1(x) = 1$, $f_2(x) = x_1$, ..., $f_n(x) = x_{n-1}$
- Therefore, the linear regression as data fitting model is defined as

$$\hat{y} = \hat{f}(x) = \theta_1 + \theta_2 x_1 + \theta_3 x_2 + \cdots + \theta_n x_{n-1}, \quad \text{where } \theta_1 = v \text{ and } \theta_i = \beta_{i-1} \text{ for } i = 2, \dots, n$$

General data fitting model as linear regression

- **General data fitting model:**

$$\hat{f}(x) = \theta_1 f_1(x) + \theta_2 f_2(x) + \cdots + \theta_n f_n(x)$$



- Equivalent **linear regression model:**

$$\hat{y} = \hat{f}(x) = \tilde{x}^T \beta + v = x_1 \beta_1 + x_2 \beta_2 + \cdots + x_{n-1} \beta_{n-1} + v$$

- **Where we define...**

- $f_1(x) = 1 \rightarrow$ common assumption
- $\tilde{x} = (f_2(x), \dots, f_n(x))^T \rightarrow$ transformed features
- $v = \theta_1$ and $\beta_{i-1} = \theta_i$ for $i = 2, \dots, n$

Remark: Our general data fitting framework is nothing more than linear regression on transformed data

Validation

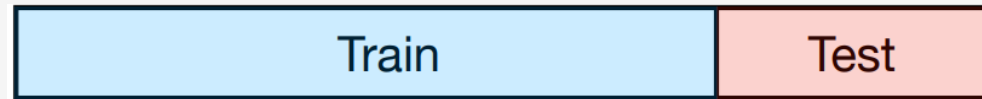
Model Generalization

- The **main goal** of a model is to predict the outcome of **new (unseen) data**
 - **Not** to predict the outcome of known data (used in the learning process)
- We say that a model **generalizes** if it makes reasonable predictions on **new data**
- A model that makes accurate predictions on known data and poor predictions on new data is said to suffer from **overfitting**
- **Always remember:** the objective function (i.e., the training error) is just a "**surrogate**" of the true (unknown) goal.



Model Validation

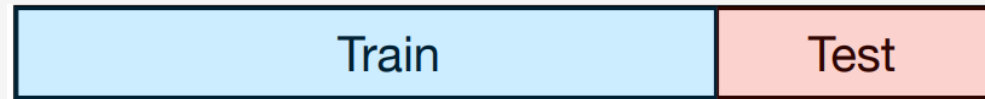
How do we **assess the validity of a predictive model**?



1. Split **original data** in **training** and **tests** sets (e.g., 80%-20% or 90%-10%)
2. Compute the θ^* (train) of the model using the **training set only**
3. Check model's prediction accuracy (effectiveness) on the **test set**
 - a) Compare the MSE prediction error on **training set** and **test set** (**train error –vs- test error**)
 - b) If they are **similar**, then the model will **generalize** on unseen (new) data points

Validation of many models

What happens when we have trained many different models?



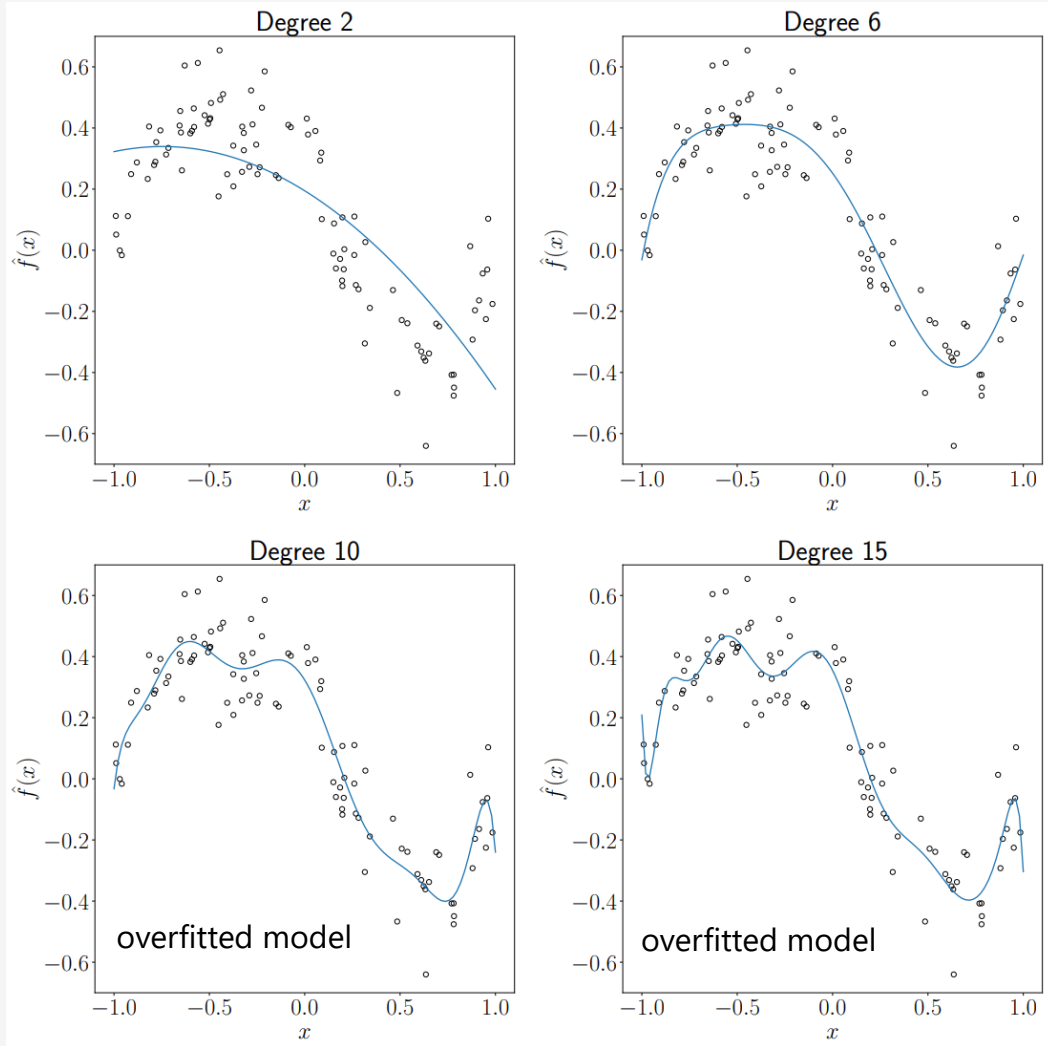
Model #1: Polynomials of different degrees

Model #2: Uses different transformed features

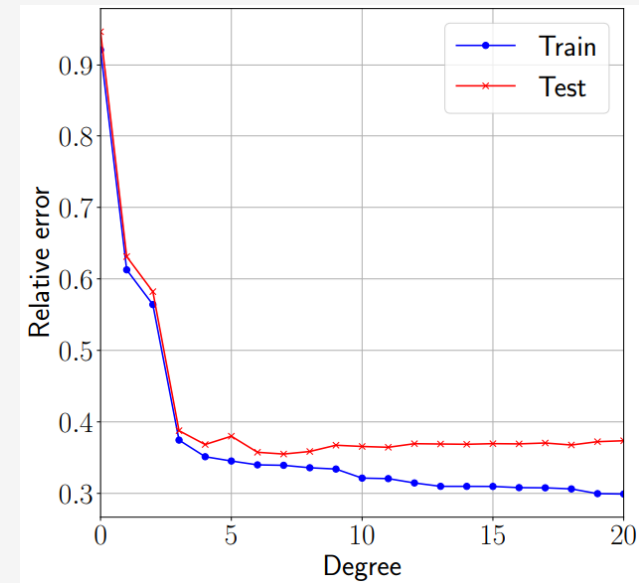
In practice: Choose the model with the lowest test error

Example 1

Example – polynomial fit



100 points in the training set
10 points in the test set



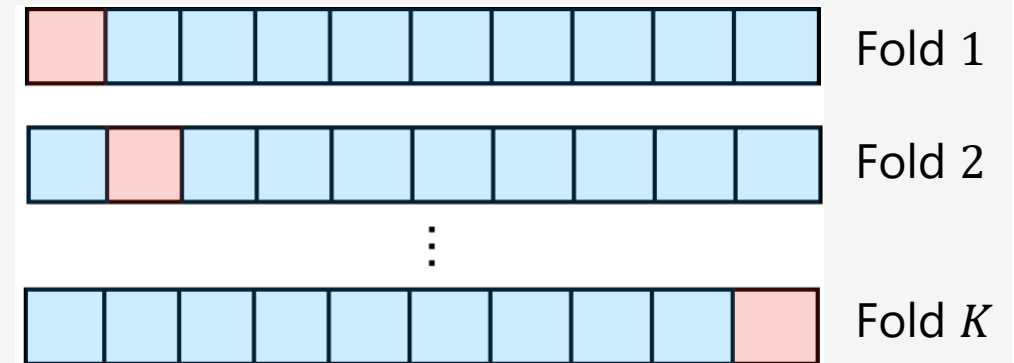
Polynomial models with degree 6 or 7 or 8 seem to perform better than others

Because: their training and test MSE's are small enough, and they are close to each other

Cross Validation (CV)

- The main goal of CV is to **check which method provides the better models**
- **Examples of methods:** basis functions ($f_i(x)$), or the choice of features
- **Main steps:**

1. Divide data in K folds
2. For $i = 1, \dots, 10$
 - a) **train your model** using all folds **except** i
 - b) **test your model** on data from fold i



Remark: this process results in **many (similar) models**.

❑ To get the final model:

- We train over the **whole dataset**! If $\text{trainMSE} \approx \text{testMSE}$, then the model **generalizes**
- We take the **average** of the parameters obtained from each fold training

Example 2

House price prediction

Feature engineering



- We have data from **774 house sales** in Sacramento area

Model: $\hat{f}(x) = \theta_1 f_1(x) + \dots + \theta_8 f_8(x)$

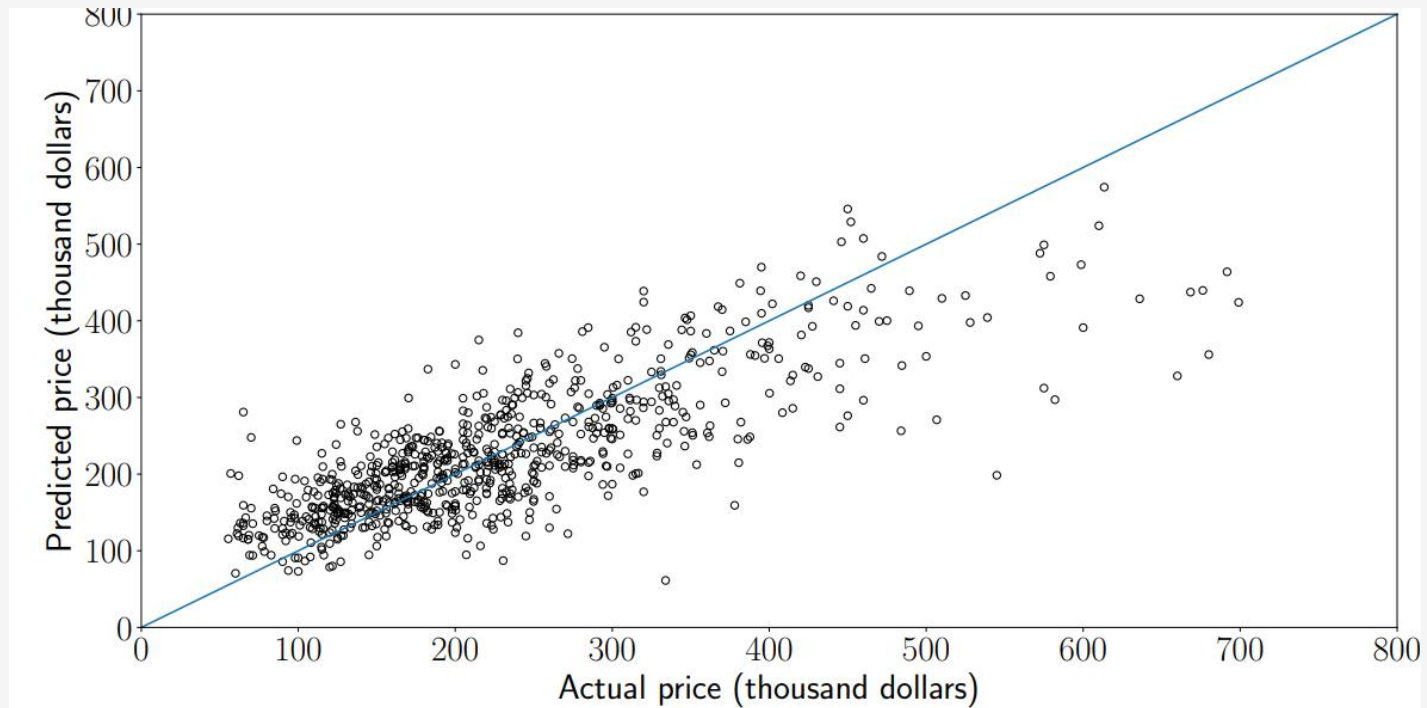
Basic features:

- x_1 : area of the house
- x_2 : number of bedrooms
- x_3 : 1, if condo; 0, otherwise
- x_4 : ZIP code

Transformed features:

- $f_1(x) = 1$ (offset)
- $f_2(x) = x_1$
- $f_3(x) = \max\{0, x_1 - 1.5\}$ (is the house area more than $1,500 ft^2$)
- $f_4(x) = x_2$
- $f_5(x) = x_3$
- $f_6(x), f_7(x), f_8(x)$: Boolean functions of x_4 to encode 4 groups
Nearby ZIP codes (i.e., neighborhood)

Fitted model over the entire dataset



How can we be sure the model generalizes?

- **5-fold cross validation** ($774/5 \approx 155$)

Fold	Train error	Test error	θ_1	θ_2	θ_3	θ_4	θ_5	θ_6	θ_7	θ_8
1	0.26	0.30	115.60	177.53	-47.05	-14.64	-13.90	-112.00	-122.59	-36.51
2	0.26	0.29	121.59	165.48	-30.35	-17.74	-20.21	-95.22	-103.67	-7.94
3	0.27	0.25	117.83	181.18	-49.78	-19.30	-18.68	-106.50	-112.76	-32.37
4	0.27	0.25	104.00	174.54	-41.96	-18.81	-17.13	-85.68	-92.09	-6.99
5	0.27	0.27	119.40	178.57	-44.82	-19.50	-25.98	-103.95	-111.56	-36.89

Good feature choices because:

1. The model parameters reasonably stable across all folds
2. Similar training and test errors (generalizes well)

References

- S. Boyd, L. Vandenberghe: Introduction to Applied Linear Algebra – Vectors, Matrices, and Least Squares
- Chapter 13: Least squares data fitting

Next lecture

- Multi-objective least squares
